

REPUBLIC OF CAMEROON

Peace – Work – Fatherland



AFRICAN INSTITUTE OF COMPUTER SCIENCES–
CAMEROON PAUL BIYA TECHNOLOGICAL CENTER
OF EXCELLENCE

Email : contact@iaicameroun.com



AFRICAN INSTITUTE OF COMPUTER SCIENCES–
CAMEROON PAUL BIYA TECHNOLOGICAL CENTER
OF EXCELLENCE

Email : contact@iaicameroun.com

BIG DATA PROJECT

THEME: GRAPH DATABASES

Submitted by:

GROUP 8

Level III Students at AICS Cameroon

COURSE INSTRUCTOR

Mme. NDIFOR ROGER

2025 – 2026

Academic Year

SE3B

WELCOME TO OUR PRESENTATION

BY GROUP 8



GROUP MEMBERS

- ❑ NGANG MARTIN FRUFOBOZI (19.5%)
 - ❑ NOUBISSIE SOH DRIANE LAURIE (21%)
 - ❑ ONANA JACQUE MARIE MICHEL ARCHANGE (20%)
 - ❑ SINOUE TAKOUGOUM NGUELE INES (19.5%)
 - ❑ SOMO REAN GIGGIS KINYUY (20% - Group Leader)
- 
- 
- 
- 

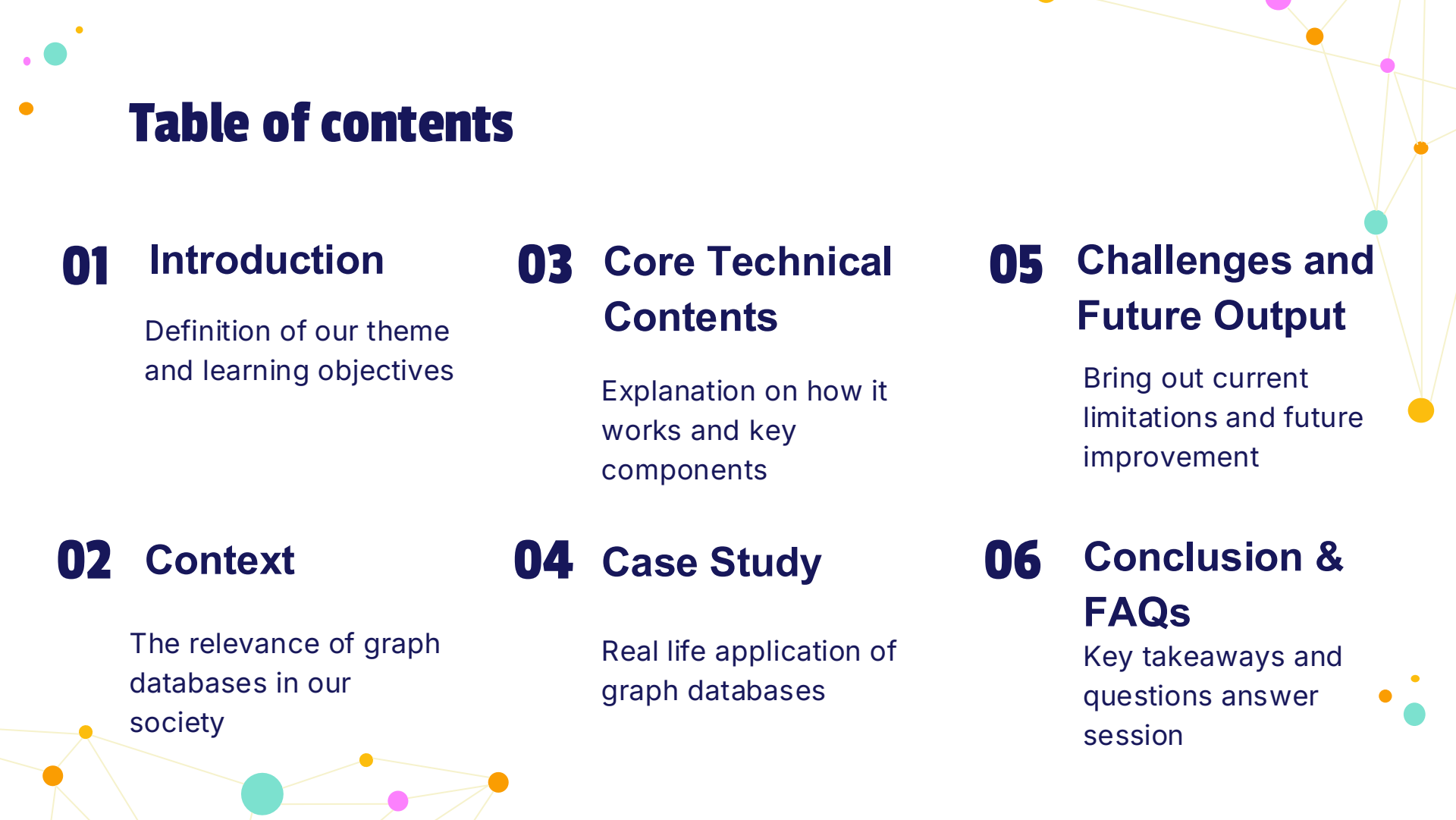


Table of contents

01 Introduction

Definition of our theme and learning objectives

03 Core Technical Contents

Explanation on how it works and key components

05 Challenges and Future Output

Bring out current limitations and future improvement

02 Context

The relevance of graph databases in our society

04 Case Study

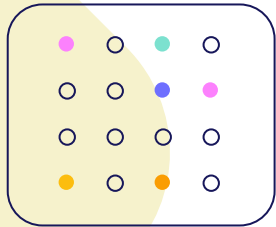
Real life application of graph databases

06 Conclusion & FAQs

Key takeaways and questions answer session

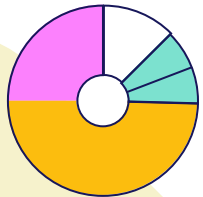
01.

INTRODUCTION



Introduction

In today's digital world, organizations generate and collect massive amounts of data every second from social media, e-commerce platforms, banking systems, IoT devices, and online services. This data is not only large in volume, but also highly connected. For example, users interact with other users, purchase products, share content, and create networks of relationships. Traditional databases often struggle to manage and analyze these complex connections efficiently. This is why graph databases have become an important solution in the era of Big Data, providing a faster and more flexible way to store and explore connected information.





A **graph database** is a type of database that stores data in the form of a **graph**, rather than in traditional tables. In a graph database, the data is represented using **nodes**, **relationships (edges)**, and **properties**



What will we learn in this lesson?



01.

Conceptual Understanding

Understand what graph databases are.
Identify key components such as nodes, relationships, and properties

02.

Big Data Relevance

Explain the importance of graph databases in Big Data environments

03.

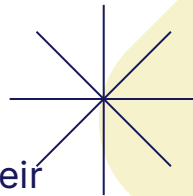
Real life applications

Recognize real-world use cases of graph databases

04.

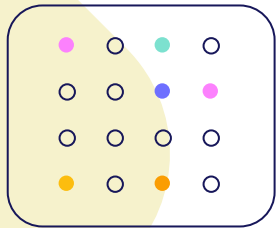
Comparative Analysis

Compare graph databases with other database models and understand their strengths and limitations



02.

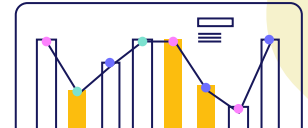
CONTEXT





Why Graph Databases?

- **Data Has Changed**
- Modern data is **highly connected**, not isolated
- Users, devices, transactions, and events form **networks of relationships**
- Understanding **how data is connected**



Traditional Challenges

Traditional data systems struggle to manage Big Data due to issues of scale, complexity, and speed. Some of them include:



**How
many?**

Volume



**How
different?**

Variety



**How
fast?**

Velocity



**How
fragmented?**

Data Silos



**How
Costly?**

Infrastructure
Cost

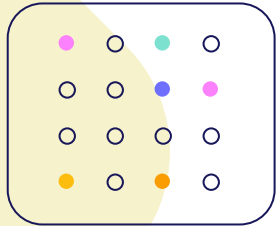


**How
accurate?**

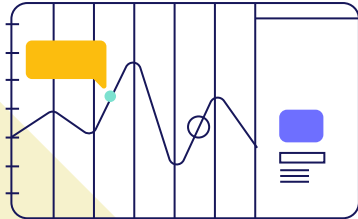
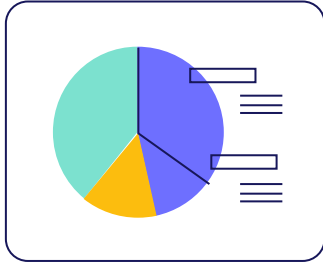
Data
Quality

03.

CORE TECHNICAL CONTENTS



Mathematical concepts



Numbers

Math is all about numbers, like counting how many friends you have or how many apples are in a basket

Shapes

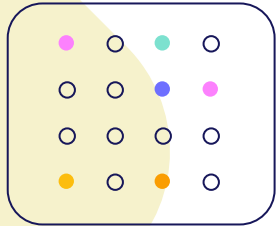
Let's talk about different shapes. Think of a square, a circle, or a triangle — they help us understand

Adding and taking away

We use math to add things together, like counting toys, and to take things away

04.

CASE STUDY



◆ Case Study: Facebook

Problems Faced

Euclid publishes "Elements"

Billions of Users

Gödel's Incompleteness Theorem

**Real-time
Recommendations**

Complex Relationships

The development of calculus

**Slow Joins in
Relational DBS**

Proof of Fermat's Last Theorem

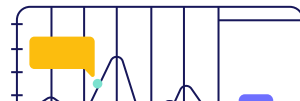


Why the choice of graph databases?

Give students cards and challenge them to build the tallest tower within a time limit and measure the height. Afterwards you can reflect on how to apply math concepts to the experiment



1. Natural representation of relationships
2. Faster Traversal
3. Scalable and Flexible



How Facebook Uses Graph DBs



Users as nodes

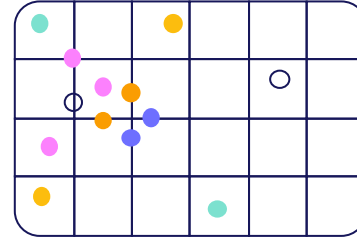
Friends as edges

Likes, comments, groups as relationships

Aspects Of Using Graph DBs

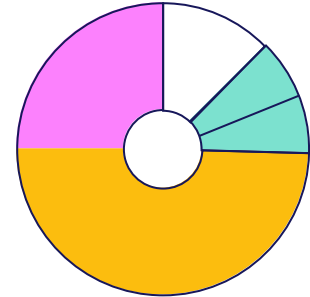
Friend Suggestions

- ❑ Friends of friends
- ❑ Same interests
- ❑ Same Location
- ❑ Computed in real time



News Feed Optimization

- ❑ Track interactions
- ❑ Prioritize close connections
- ❑ Improve relevance





Benefits Achieved and Technologies Used

Benefits



- ☐ High performance
 - ☐ Accurate Recommendations
 - ☐ Better User Experience
- 

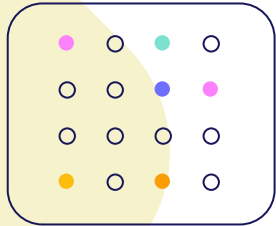


Technologies

- ☐ TAO (Facebook)
 - ☐ Neo4j
 - ☐ Amazon Neptune
- 
- 

05.

CHALLENGES AND FUTUR OUTPUT



CHALLENGES

Scalability In Big
Data Environment

1

2

High Memory and
Resource
Consumption

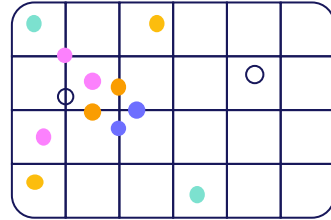
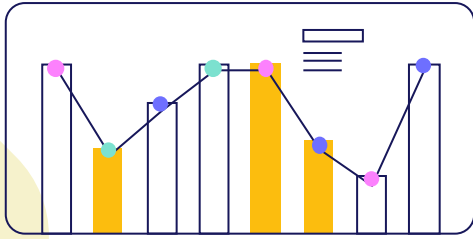
limits

Lack of
Standardization

3

4

Limited Skills and
Adoption



Futur Output

What are the futur expectations?

Futur Output

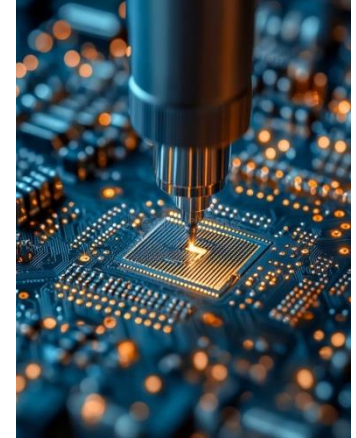
In the future we have as following expectations

**Real Time Decision
Making**

**Smart Artificial
Intelligence**

**Better Recommendation
Systems**

**New Big Data
Application**



06. Conclusion



Conclusion



Final Quotes

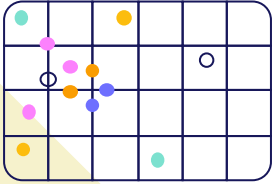
Big Data drives innovation but brings challenges like privacy, bias, and cost. Ethical, problem-focused analytics and AI integration will shape its future impact.



Take home Message

Big Data turns massive challenges into opportunities. Traditional tools can't handle today's data volumes, so scalable solutions like Hadoop, Spark, and NoSQL are essential.





Thanks!

Do you have any questions?
group8@gmail.com

